

HARSHIT RASTOGI

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PROFESSIONAL SUMMARY

Backend Engineer with 3.5 years of experience building scalable, cloud-native enterprise platforms using Python, FastAPI, and AWS (ECS, Fargate, Lambda, Aurora PostgreSQL, S3). Proven experience developing AI-powered backend solutions with Amazon Bedrock, RAG, and LLM orchestration to automate enterprise workflows.

TECHNICAL SKILLS

Languages: Python, SQL, C++

Frameworks: Backend: FastAPI, Mangum, Pydantic, SQLAlchemy 2.0, LangChain, LangGraph

Cloud & Infra: AWS Lambda, Amazon ECS, AWS Fargate, Amazon Bedrock, API Gateway, Amazon S3, Aurora Serverless PostgreSQL, Amazon EventBridge Scheduler, IAM, AWS SSO

AI/ML: Claude Sonnet 4, Amazon Nova Lite, Titan Embeddings, RAG, pgvector, Prompt Engineering, MCP

DevOps & Tools: Docker, TeamCity, Liquibase, Git, Postman, pytest, boto3

PROFESSIONAL EXPERIENCE

Backend Engineer — ZS Associates, India

Aug 2023 – Present

Extension Lifecycle Manager

- Architected a serverless developer platform using **Python FastAPI + AWS Lambda** (Mangum) enabling enterprise teams to manage extension lifecycles through a 6-phase gated workflow with AI-powered automation.
- Engineered a multi-model AI orchestration layer integrating **Amazon Bedrock** (Nova Lite, Claude, Titan Embeddings) to auto-generate dev specs, test scripts, and security audits — reducing documentation effort from ~40 hours to under 30 minutes.
- Built a custom **RAG pipeline** using pgvector and semantic chunking for context-aware AI generation from uploaded PRDs, without external RAG frameworks.
- Developed security middleware with sliding-window rate limiting, OWASP-compliant headers, prompt injection validation, and **4-tier RBAC** — all without external dependencies.

Quota Allocation & Refinement Engine

- Built a hierarchical quota cascade engine using **Decimal + Largest Remainder Method**, guaranteeing zero floating-point drift across 10-level geographic hierarchies with 10,000+ nodes.
- Implemented a configurable business rule engine with **30+ runtime-toggable validation rules** (threshold limits, zero-sum enforcement) enabling per-group customization without code deploys.
- Optimized bulk operations with 500-record batch processing and SQL pre-filtering on JSON columns, achieving **sub-second queries** on 10K+ node hierarchies.

Performance Management Platform (MBO)

Designed a multi-phase workflow engine with 4 phases (Define, Track, Self-Rate, Rate), each with independent activation logic and approval chains across a **3-tier management hierarchy**.

- Built a bidirectional **S3 data integration pipeline** for bulk import/export of user rosters and geography mappings with multi-step referential integrity validation.

Backend Engineer Intern — ZS Associates, India

Jan 2023 – Jul 2023

- Implemented **OAuth 2.0 authentication** with JWT access/refresh tokens and hierarchical RBAC for an enterprise identity management platform.

HACKATHON — 1ST PLACE WINNER

AI-Powered Platform Automation Agent | LangChain, LangGraph, Claude, Bedrock, MCP

- Architected a **4-layer agentic AI system** (Agent → Tools → Service → Actions) using LangChain/LangGraph and Claude on Bedrock to convert natural language into autonomous platform API calls.
- Built a domain-specific script linting engine and a **Chrome Extension** with side-panel UI providing contextual AI assistance via background service worker message passing.

Achievements

- Received the "AI Bridge to the Future" Award at the ZS PPS Rewards & Recognition H1'2026 for excellence in AI-powered platform automation.

EDUCATION

Bachelor of Technology in Computer Science

DIT University, India | CGPA: 8.2 / 10

2019 – 2023